

Dr. Jyoti Panda

CONTACT INFORMATION	Mechanical Engineering Department University of North Carolina, Charlotte North Carolina, USA	Phone: +1-(551) 260-1943 Email: jppanda.iit@gmail.com Homepage: https://dr-jppanda.github.io
PROFESSIONAL SUMMARY	Accomplished computational thermo-machanical modeling researcher with strong analytical, organizational, and project management skills. Demonstrated expertise in developing high-fidelity computational, experimental and data-driven models for heat exchanger, microelectronic, aerospace and mechanical systems involving fluid flow and heat transfer.	
RESEARCH INTERESTS	Computational modeling and simulation, Fluid mechanics, Heat Transfer, Computational Fluid Dynamics, Thermal management, Experiments, Microelectronic heat transfer, Heat exchangers, Thermo-electrics, Joule Heating, Fluid mixing, Acoustics, Microfluidics, Rarefied gas flows, Turbulent Flows, Data-driven Modeling, Machine Learning	
EXPERIENCE	University of Cincinnati-UNC Charlotte, USA May 2025 to Present Post-Doctoral Research Fellow Funding Agency: NASA, NSF • Conducted heat exchanger experiments to collect acoustic and thermofluidic data under varying operating conditions. • Developed machine learning models to predict heat transfer characteristics from acoustic signals. • Validated predictions against experiments, improving understanding of condensation dynamics. University of Notre Dame, USA Oct 2023 to May 2025 Post-Doctoral Research Associate Funding Agency: Defense Advanced Research Projects Agency (DARPA), USA • Conducted TCAD simulations to model Joule heating and performed thermo-electric modeling of transistors in microelectronic circuits • Predicted the thermal field in large FinFET transistor arrays using both JAX-based BTE and Fourier simulations. Research Center for Aircraft Core Technology, S. Korea Aug 2022 to Sep 2023 Senior Researcher Funding Agency: NRF-South Korea, AOARD-AFSOR-USA • Performed DSMC simulations of rarefied gas flows using SPARTA and Bird's code. • Conducted DNS simulations of decaying compressible turbulence and shock-turbulence interactions. • Investigated shock-boundary layer interactions on high-performance computing servers running Linux Indian Institute of Technology Kharagpur, India Mar 2019 to June 2021 Research Associate Funding Agency: NRB, Defense Research and Development Organisation, India • Conducted CFD simulations in Ansys Fluent to investigate drag reduction strategies for underwater vehicles using micro-dimples and grooves. • Analyzed flow behavior to quantify the impact of surface modifications on drag and hydrodynamic performance. • Implemented machine learning models to enable fast and accurate prediction of the flow field, reducing computational time.	

EDUCATION	Indian Institute of Technology Kharagpur , Kharagpur, WB, India	
	Ph.D., Computational Fluid Dynamics (Turbulence Modeling), Aug 2019	
	Thesis Topic: <i>Pressure Strain Correlation Modeling for Turbulent Flows</i>	
	Indian Institute of Engineering Science and Technology, Shibpur , WB, India	
	M.E., Mechanics of Fluids (CFD-Microfluidics), June 2015 - Department: Aerospace Engineering and Applied Mechanics - Thesis Topic: <i>Electroosmotic Mixing and Joule Heating in Microchannels</i>	
	National Institute of Electronics and Information Technology, India	
	P.G. (Post Graduate), Data Engineering, Online Course, Nov 2023	
	- Configuring Linux Platform for Data Engineering - Data Analytics and Machine Learning using Tensorflow and Keras - Big Data Analytics with Hadoop and Apache Spark	
	Biju Patnaik University of Technology, Rourkela , ODISHA, India	
	B.Tech., Mechanical Engineering, June 2012 - Project: <i>Design and fabrication of a pneumatic material handling system</i> - CGPA 8/10	
PROGRAMMING AND SOFTWARE SKILLS	Programming: Python, C/C++, Fortran Software/Codes: Docker and Singularity container ANACONDA (For python environments) JAX-Fluids (DNS and LES) (Parallel CUDA, Python) STREAMS (DNS) (Parallel MPI/CUDA, FORTRAN) GiftBTE (Boltzmann transport equation) (submicron thermal transport) OpenFOAM (RANS, LES, DSMC) (Parallel MPI, C++) SU2 (Parallel MPI, C++) ANSYS Fluent (RANS and RSTM) Gmsh/Ansys-workbench/Gambit (Meshing) SPARTA and Prof. Bird's code (DSMC) TensorFlow and Keras (Deep learning) TensorFlowFoam(Linking neural network models with OpenFoam) Scikit Learn (Machine Learning), MATLAB	
EXPERIMENTAL WORK	Instrument: Acoustic Doppler Velocimeter Principle of operation: Doppler Shift Measured parameters: Three fluctuating turbulent velocity components in grid generated turbulence with and without mean strain Location: Recirculating water tank, Ship Hydrodynamics Lab, IIT Kharagpur	
SCHOLARSHIPS	- MHRD government of India fellowship for doctoral studies , India - MHRD government of India fellowship for PG studies , India	2015-2018 2013-2015

1. **Panda, J.**, Shang, W., Liu, Y., Zhou, J., Liang, J., Wang, J.X. and Luo, T.,“Scalable non-diffusive thermal analysis of field-effect transistors using neural operators” 2026, *APL Electronic Devices*, 2(1).
2. **Panda, J.**, Warrior, H.,“Numerical studies on drag reduction of an axisymmetric body of revolution with antiturbulence surface” 2021, *ASME Journal of Offshore Mechanics and Arctic Engineering*, 143(6), p.064501. (Impact Factor: 1.355) (Q2 Energy)
3. Jhou, J, Shang, W, Li, Ruiyang, Liu,Yi ,**Panda, J.**,...,Wang, J , Luo, T,“Physics-Informed Neural Networks with Hard-Encoded Angle-Dependent Boundary Conditions for Phonon Boltzmann Transport Equation”2025, *Materials Today Physics*, 59, pp.101922.(Q1 Energy)
4. Shang, W, Jhou, J,**Panda, J.**,Wang, J , Luo, T,“JAX-BTE: a GPU-accelerated differentiable solver for phonon Boltzmann transport equations”2025, *npj Computational Materials*, 11(1), pp.1-12.(Q1 Computer Science Applications)
5. **Panda, J.**, Warrior, H.,“Modeling pressure strain correlation for turbulent flows using deep neural networks” 2021, *Proceedings of the Institution of mechanical engineers, Part C: Journal of Mechanical Engg. Science.* (Impact Factor: 1.762). (Q2 Mechanical Engineering)
6. **Panda, J.**, Warrior, H.,“Data-driven prediction of complex flow field over an axisymmetric body of revolution using Machine Learning” 2022, *ASME Journal of Offshore Mechanics and Arctic Engineering.* (Impact Factor: 1.355) (Q2 Energy)
7. Mitra, A. **Panda, J.**, Warrior, H.,“The effects of free stream turbulence on the hydrodynamic characteristics of an AUV hull form” 2019. *Ocean Engineering, Vol. 174 (2) / 148-158.* (Impact Factor: 3.795) (Q1 Environmental engineering)
8. Mitra, A., **Panda, J.**, Warrior, H.,“Experimental and numerical investigation of the hydrodynamic characteristics of Autonomous Underwater Vehicles over seabeds with complex topography”2020. *Ocean Engineering, Volume 198, 106978 .* (Impact Factor: 3.795) (Q1 Environmental engineering)
9. **Panda, J.**, Warrior, H.,“A representation theory based model for the rapid pressure strain correlation of turbulence”2018, *ASME Journal of Fluids Engg.*, Vol. 140 / 081101-1. (Impact Factor: 1.995) (Q2 Mechanical Engineering)
10. **Panda, J.**, Warrior, H.,“Evaluation of machine learning algorithms for predictive Reynolds stress transport modeling” 2021, *Acta Mechanica Sinica* , (Impact Factor: 1.975) (Q2 Computational Mechanics)
11. **Panda, J.**, Kumar, B., Kumar, A.,Patil, A.,“Influence of twisted tape length on the thermal performance of a heat exchanger tube” 2022, *Numerical Heat Transfer, Part A: Applications*, (Impact Factor: 2.928) (Q2 Condensed matter physics)
12. **Panda, J.**, Kumar, B., Patil A., Kumar M. “Machine learning assisted modelling of thermohydraulic correlations for heat exchangers with twisted tape inserts, 2023, *Acta Mechanica Sinica* (Accepted), (Impact Factor: 1.975) (Q2 Computational Mechanics)
13. **Panda, J.**, Warrior, H., Maity, S., Mitra, A.,Sasmal, K.,“An improved model including length scale anisotropy for the pressure strain correlation of turbulence” 2017, *ASME Journal of Fluids Engineering*, Vol. 139 / 044503-1. (Impact Factor: 1.995) (Q2 Mechanical Engineering)

14. **Panda, J.**, “A review of pressure strain correlation modeling for Reynolds stress models ” 2019, *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*. DOI:<https://doi.org/10.1177/095440-6219893397>. (Impact Factor: 1.762) (Q2 Mechanical Engineering)
15. **Panda, J.**, Mitra, A., Warrior, H., “A review on the hydrodynamic characteristics of autonomous underwater vehicles” 2020. *Proceedings of the Institution of Mechanical Engineers, Part M: Journal of Engineering for the Maritime Environment*, DOI: <https://doi.org/10.1177/1475090220936896>. (Impact Factor: 1.389) (Q2 Mechanical Engineering)
16. **Panda, J.**, “A reliable pressure strain correlation model for complex turbulent flows” 2020. *Journal of applied fluid mechanics, Vol. 13, No. 4, pp. 1167-1178*. (Impact Factor: 1.783) (Q3 Condensed matter physics)
17. **Panda, J.**, Sasmal, K., Maity, S., Warrior, H., “A Simple Nonlinear Eddy Viscosity Model for Geophysical Turbulent Flows” 2020, *Journal of Applied Fluid Mechanics*, 14(3). (Impact Factor: 1.783) (Q3 Condensed matter physics)
18. Das A., Das, SR., **Panda, J.**, Dey A., Gajrani KK., Somani N., Gupta N. “Machine learning based modelling and optimization in hard turning of AISI D6 steel with advanced AlTiSiN coated carbide inserts, 2022, *Surface Review and Letters*. (Impact Factor: 1.303) (Q3 Materials chemistry)
19. Somani, N., Walia, A., Gupta, N., **Panda, J.**, Das, A., Das S. “Data driven surrogate model-based optimization of the process parameters in electric discharge machining of D2 steel using Cu-SiC composite tool” 2023, *Revista de Metalurgia*. (Impact Factor: 0.8) (Q3 Metals and Alloys)
20. **Panda, J.**, Mitra, A., Joshi, A., Warrior, H., “Experimental and numerical analysis of grid generated turbulence with and without mean strain” 2018, *Experimental Thermal and Fluid Science, Vol. 98 (11) / 594-603*. (Impact Factor: 3.232) (Q1 Aerospace engineering)
21. **Panda, J.**, Handique J., Warrior, H., “Mechanics of drag reduction of an axisymmetric body of revolution with shallow dimples” 2022, *Proceedings of the iMech, Part M: Journal of Engineering for Maritime Environment*. (Impact Factor: 1.389) (Q2 Mechanical Engineering)
22. **Panda, J.** “Machine learning for Naval Architecture Ocean and Marine Engineering” 2023, *Journal of Marine Science and Technology*. (Impact Factor: 2.005) (Q1 Mechanical Engineering)
23. Mitra, A., **Panda, J.**, Warrior, H., “The hydrodynamic characteristics of autonomous underwater vehicles in rotating flow fields” 2023, *Proceedings of the iMech, Part M: Journal of Engineering for Maritime Environment* (Q2 Mechanical Engineering).

INTERNATIONAL
CONFERENCE
PUBLICATIONS

1. **Panda, J.**, ..., Wang, J , Luo, T, “Scalable Thermal Analysis of Field-Effect Transistors Using Neural Operators” 2025. *18th U.S. National congress on computational mechanics, Chicago, Illinois, July 20-24, 2025*.
2. **Panda, J.**, ..., Wang, J , Luo, T, “Data-Driven Prediction of Thermal Field in Field-Effect Transistors Using Deep Neural Networks” 2024. *ASME 2024 Heat Transfer Summer Conference (SHTC2024)*.

3. Zhou J, **Panda, J.**,...,Wang, J , Luo, T, “ Jax-Bte: A Differentiable Hybrid Neural Solver for Deep Learning Accelerated Thermal Modeling of Nanoelectronics ” 2024. *ASME 2024 Heat Transfer Summer Conference (SHTC2024)*.
4. Shang W, **Panda, J.**,...,Wang, J , Luo, T, “Physics-Integrated Hybrid Machine Learning Model for Phonon Bte ” 2024. *ASME 2024 Heat Transfer Summer Conference (SHTC2024)* .
5. Zhou J, **Panda, J.**,...,Wang, J , Luo, T, “ Physics-Informed Neural Networks for Transistor Thermal Modeling Using Phonon Boltzmann Transport Equation ” 2024. *ASME 2024 Heat Transfer Summer Conference (SHTC2024)*.
6. J Zhou, R Li, W Shang, B Zhang, Z Xu, **Panda, J.**, J Wang, T Luo, T Beechem, E Walker, “Efficient Physics-Informed Deep Learning Model for Multiscale Thermal Analysis” 2024. *Government Microcircuit Applications and Critical Technology (GOMACTech) Conference*.
7. **Panda, J.**, Myong, R. S., “Subgrid Modeling for Large Eddy Simulation of Shock Boundary Layer Interaction Using Machine Learning” 2023. *The 14th Asian Computational Fluid Dynamics Conference, CSIR, National Aerospace Laboratories, Bengaluru, India*.
8. **Panda, J.**, Sengupta, B., Myong, R. S., “Direct numerical simulation of shock turbulence interaction with bulk viscosity effects” 2023. *The 34th International Symposium on Shock Waves, Daegu, Korea*.
9. **Panda, J.**, Gupta, S., Pal, D., “Computational Analysis of Liquid-Liquid Mixing In a T-Shaped Serially Connected Converging-Diverging microchannel” 2014. *59th Congress of ISTAM, IIT Kharagpur, India*.
10. Mohapatra, P., **Panda, J.**, Pal, D., “Electro-osmotic Flow and Mixing in a Micro-channel: A Numerical Study” 2014. *59th Congress of ISTAM, IIT Kharagpur, India*.
11. **Panda, J.**, Warrior, H., Maity, S., “Pressure Strain Correlation for decaying homogeneous turbulence” 2016. *Fluid Mechanics and Fluid Power Conference held at MNNIT Allahabad, India*.
12. Joshi, A., Warrior, H., **Panda, J.** “An Improved Single Point Closure Model Based on Dissipation Anisotropy for Geophysical Turbulent Flows” 2018. *Int. Conference on Oceanography held at Miami, USA*.
13. Gupta, S., **Panda, J.**, Nandi, N. “A Model Study of Free Vortex Flow” 2014. *ICTACEM Conference held at IIT Kharagpur, India*.

REFERENCES

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PERSONAL
PROFILE

NAME: Jyoti Prakash Panda
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